

1.6: Polar Coordinates

1. The Polar Coordinate System

In polar coordinates a point is given by (r, θ) where

- r is the distance from the pole (origin),
- θ is the angle measured anticlockwise from the initial line (positive x -axis).

OCR convention: $r \geq 0$.

Conversion Cartesian $\leftrightarrow$ Polar

$$x = r \cos \theta, \quad y = r \sin \theta$$

$$r = \sqrt{x^2 + y^2}, \quad \tan \theta = \frac{y}{x} \text{ (choose } \theta \text{ in the correct quadrant)}$$

2. Sketching Polar Curves

To sketch $r = f(\theta)$, tabulate key θ values $(0, \pi/6, \pi/4, \pi/3, \pi/2, \dots)$ and plot. Some useful standard curves:

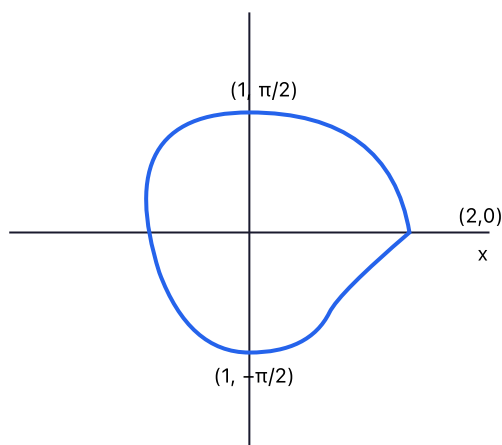
Equation	Shape
$r = a$	Circle, radius a , centre origin
$r = 2a \cos \theta$	Circle, radius a , centre $(a, 0)$

$r = 2a \sin \theta$	Circle, radius a , centre $(0, a)$
$r = a(1 + \cos \theta)$	Cardioid (heart-shaped)
$r = a(1 + k \cos \theta), \frac{1}{2} < k < 1$	Dimpled limaçon
$r = a + b \cos \theta, a < b$	Limaçon with inner loop
$r = a \cos n\theta$ or $r = a \sin n\theta$	Rose curve — n petals if odd, $2n$ if even
$r = a\theta$	Archimedean spiral
$r^2 = a^2 \cos 2\theta$	Lemniscate (figure of eight)

Worked Sketch — The Cardioid $r = 1 + \cos \theta$

Key points: $\theta = 0 \Rightarrow r = 2$; $\theta = \pi/2 \Rightarrow r = 1$; $\theta = \pi \Rightarrow r = 0$ (the pole).

Cardioid $r = 1 + \cos \theta$.



3. Tangents at the Pole

If $r = 0$ when $\theta = \alpha$, then the line $\theta = \alpha$ is tangent to the curve at the pole.

Tangents at the pole for $r = \sin 2\theta$.

$r = 0$ when $\sin 2\theta = 0$, i.e. $\theta = 0, \pi/2, \pi, 3\pi/2$. These four half-lines are the pole tangents — the curve is a four-petal rose.

4. Area Enclosed by a Polar Curve

The area bounded by $r = f(\theta)$, $\theta = \alpha$, and $\theta = \beta$ is

$$A = \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta = \frac{1}{2} \int_{\alpha}^{\beta} [f(\theta)]^2 d\theta.$$

Area enclosed by the cardioid $r = 1 + \cos \theta$.

Symmetry: compute area for $\theta \in [0, \pi]$ and double.

$$\begin{aligned} A &= \frac{1}{2} \cdot 2 \int_0^{\pi} (1 + \cos \theta)^2 d\theta = \int_0^{\pi} (1 + 2 \cos \theta + \cos^2 \theta) d\theta \\ &= \int_0^{\pi} \left(1 + 2 \cos \theta + \frac{1 + \cos 2\theta}{2}\right) d\theta \\ &= \left[\frac{3}{2}\theta + 2 \sin \theta + \frac{1}{4} \sin 2\theta\right]_0^{\pi} = \frac{3\pi}{2}. \end{aligned}$$

OCR Exam Tip: Be careful about the limits of integration — they should trace out the region *once*. For symmetric curves, use symmetry to simplify.

5. Tangents in Cartesian Form

From $x = r \cos \theta$, $y = r \sin \theta$, the gradient is:

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{\dot{r} \sin \theta + r \cos \theta}{\dot{r} \cos \theta - r \sin \theta}$$

Horizontal tangents: $dy/d\theta = 0$. Vertical tangents: $dx/d\theta = 0$.

6. Areas Between Two Polar Curves

For the region inside $r = g(\theta)$ and outside $r = f(\theta)$, with $g \geq f$ on $[\alpha, \beta]$:

$$A = \frac{1}{2} \int_{\alpha}^{\beta} (g(\theta)^2 - f(\theta)^2) d\theta.$$

Find intersection points by solving $f(\theta) = g(\theta)$ — remember to consider the pole separately.

7. Exam Style Strategy

- Always sketch first — it reveals symmetry and tells you whether the region is traced once.
- State your limits carefully (e.g. "for $-\pi \leq \theta \leq \pi$ the full cardioid is covered").
- For "area between two curves", find all intersection angles, including the pole.
- Standard integrals you will need: $\int \cos^2 \theta d\theta$, $\int \sin^2 \theta d\theta$, $\int \cos n\theta d\theta$ — memorise or derive via double-angle formulas.